

8	(a) two (light) nuclei combine to form a more massive nucleus	M1	A1	[2]
(b) (i)	$\Delta m = (2.01410 \text{ u} + 1.00728 \text{ u}) - 3.01605 \text{ u}$ $= 5.33 \times 10^{-3} \text{ u}$ $\text{energy} = c^2 \times \Delta m$ $= 5.33 \times 10^{-3} \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2$ $= 8.0 \times 10^{-13} \text{ J}$	C1	C1	[3]
(ii)	speed/kinetic energy of proton and deuterium must be very large so that the nuclei can overcome electrostatic repulsion	B1	B1	[2]

Section B